

Use Foil in Walls?



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Q. I'm desperate for some balanced information. I'm putting new cement-based clapboard siding on my house on top of a plywood shell. The siding contractor said that he typically wraps the house in Tyvek. I live in the southwest corner of the Mojave Desert at about 3,200 ft above sea level and here we have both hot and cold seasons. We can go down to the teens in the winter, and at times in the summer it hits 120°F. The climate is typically dry, but it can pour during the winter storm months. I want to do something to increase the heat-and-cool retention. I understand that Tyvek does not do this. I thought that there might be some thin insulative medium that we could apply under the painted cement siding. I researched on your Web site and found no mention of a material that I found at savenrg.com (www.savenrg.com). This is a reflective film to be installed in double layers and separated from the siding with a 1/4-inch gap. The site touts the improved heat-and-cool retention provided by this material, which is called a radiant reflector. It is apparently not porous, and after reading some about vapor barriers on your site, I have begun to wonder about the effects of wrapping the house in this film if it needs to breathe in some way. I hope that you can offer some information that I can use to make an informed decision.

—Wandering In the Desert



SupeRwrap, shown above, is a product of Horizon Energy Systems.

A. Putting a vapor barrier on the outside of a structure (outside of the insulation) can be a problem when it is cold outside and significant moisture is generated inside the structure. The vapor can penetrate the structure from the inside and can condense on the insulation inside the wall. In the desert climate, moisture is less of a problem than in very cold or humid climates. Good information on vapor barriers can be found at the Building Science Corporation Web site, www.buildingscience.com/houses-thatwork/default.htm.

At the Web site you mentioned, it says that climates with cold winters should not use this product on the outside of the insulation. In cooling-dominant climates with mild winters, it may be fine, depending on how cold it gets outside, how much moisture is generated inside, how well exhausted that moisture is (especially from the bathroom and kitchen areas), and in general how well ventilated the house is. The kinds of indoor finish materials and coatings that are used can also matter. If all of these were addressed so as to minimize the amount of water vapor making its way into the wall, in your climate it should be all right to use

the product described on the site. The site is correct that an air space must be maintained for the radiant barrier to be effective. I assume the 1/4-inch gap between the reflective film and the siding would be provided by the use of lath or furring.

In any case, a barrier to liquid water is needed between the siding and the structure. Tyvek works in the same manner as a number of materials that are used for breathable but water-resistant clothing, such as GoreTex. Such materials have microscopic holes in them that allow water vapor (individual molecules of water) to pass through but that block liquid water (many molecules bound together and therefore too big to make it through the pores in the material).

As far as resistance to conductive heat flow is concerned, you could also install rigid foam sheathing, which comes in thicknesses down to 1/4 inch. A typical 1/2-inch-thick rigid foam product provides R-3, for example. But if you have the space, a thicker product will provide more thermal resistance. Check with the manufacturer of any proposed insulation product to see if it can serve as a liquid barrier, or whether an additional barrier product is needed.

